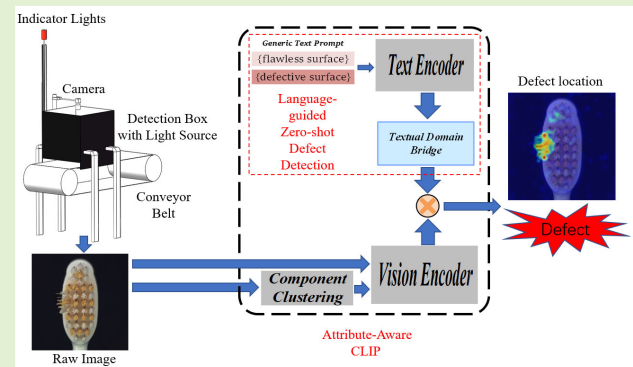


Zero-Shot Defect Detection With Anomaly Attribute Awareness via Textual Domain Bridge

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Abstract—Visual defect detection is crucial for industrial quality control in intelligent manufacturing. Previous research requires target-specific data to train the model for each inspection task. However, due to the challenges of collecting proprietary data and model-training time costs, zero-shot defect detection (ZSDD) has become an emerging topic in the field. ZSDD, which requires models trained with auxiliary data, can detect defects on different products without target-data training. Recently, large pre-trained vision-language models (VLMs), such as contrastive language-image pre-training model (CLIP), have demonstrated revolutionary generality with competitive zero-shot performance across various downstream tasks. However, VLMs have limitations in defect detection, which are designed to focus on identifying category semantics of the objects rather than sensing object attributes (defective/nondefective). The current VLMs-based ZSDD methods require manually crafted text prompts to guide the discovery of anomaly attributes. In this article, we propose a novel ZSDD method, namely attribute-aware CLIP, to adapt CLIP for anomaly attribute discovery without designing specific textual prompts. The core is designing a textual domain bridge, which transforms simple general textual prompt features into prompt embeddings better aligned with the attribute awareness. This enables the model to perceive the attributes of objects by text–image feature matching, bridging the gap between object semantic recognition and attribute discovery. Additionally, we perform component clustering on the images to break down the overall object semantics, encouraging the model to focus on attribute awareness. Extensive experiments on 16 real-world defect datasets demonstrate that our method achieves state-of-the-art (SOTA) ZSDD performance in diverse class-semantic datasets.

Index Terms—Attribute awareness, component clustering, defect detection, intelligent manufacturing, textual domain bridge, vision-language model (VLM), zero-shot.



I. INTRODUCTION

VISUAL defect detection plays a pivotal role in industrial quality control of intelligent manufacturing [1]. Compared to the high labor-cost and inevitable fatigue of manual inspection, visual defect detection is more efficient and reliable. Consequently, automated optical inspection (AOI) equipment is widely used in the manufacturing of various

industrial products, such as rail surface [2], chip package [3], printed circuit board (PCB) [4], and tire mold [5]. Traditional vision defect detection methods focus on type-specific products, requiring corresponding feature extraction techniques and meticulously crafted detection operators [6], [7], [8]. Such intricate tasks require extensive designing time and expensive customization costs. In recent years, deep learning-based visual defect detection has been extensively studied, encompassing both supervised learning methods [9], [10], [11] and unsupervised anomaly detection methods [12], [13], [14], [15], [16]. The significant region extraction (SF)-YOLO metal gear end-face defect detection method [17] is proposed to inspect the inhomogeneity structure and the multiscale, small-sized defects on the gear end-face. Uncertainty-aware unsupervised method [18] based on reconstructive and discriminative models is proposed for PCB defect detection. These data-driven detection methods allow for the development of specific detection algorithms by feeding sufficient training data into the model, effectively addressing the issue of algorithm customization.

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However, these data-driven methods require the collection of ample training data. Visual defect detection is a long-tail problem, where normal samples make up the vast majority of the total samples and defects are rare with a wide range of variations. Especially in some industrial scenarios (e.g., nuclear fuel rod manufacturing), these image samples are often considered private data. In addition, the trained deep-learning methods are only applicable to the detection of product categories according to the training data. They cannot quickly adapt to unseen products or products with significant intraclass variability. This limitation severely hampers production line efficiency, especially in flexible manufacturing tailored to customer needs. Specifically, due to the varying customer demands and the high level of product customization, different products often need to be manufactured on the same production line. For example, plain masks and printed masks differ only in the raw materials used, while the rest of the production processes are largely identical. Waiting for the collection of specific data and data-driven model training leads to frequent production halts. This not only reduces production efficiency but also significantly increases production costs.

The above challenge motivates our interest in zero-shot defect detection (ZSDD), which is an emerging task for the defect detection community. The ZSDD task requires delivering a frozen model trained with auxiliary data to the user that can detect defects without any target data for retraining, i.e., cold-start detection for unseen data. Due to the diverse visual appearance of defects and the inspected targets, accurate ZSDD is clearly a highly challenging task. Recently, large pretrained vision-language models (VLMs) [19], [20], [21] have presented strong zero-shot recognition ability in various visual tasks. Using annotated text–image pairs for large-scale training, the VLMs learn broad semantic representations. Relevant information can be extracted from these models using text prompts without additional fine-tuning in the target domain, enabling zero-shot transfer to downstream tasks such as image classification [20] and object detection [22]. Among these VLMs, contrastive language-image pre-training model (CLIP) [20], which is pretrained on billions of text–image pairs, is the most representative and widely applied as a foundational model for downstream tasks. The defect detection community has also taken note of CLIP’s powerful zero-shot generalization capabilities. WinCLIP [23] is a seminal work in the ZSDD task, where a large number of handcrafted text prompts were designed to leverage CLIP’s text–image alignment capabilities to inspect defects. Subsequent ZSDD works [24], [25] also followed a similar technical path, requiring the design of numerous suitable text description prompts to inject additional expert knowledge into the CLIP model [24].

Obviously, the requirement for experts to craft numerous and meticulously designed text prompts for the ZSDD paradigm greatly increases the preparation time and economic cost of using the model. This is because the ZSDD task inherently differs from other zero-shot visual tasks, i.e., task domain gap. As shown in Fig. 1, the design and training data of the CLIP model focus on the category semantic of the foreground object rather than its attributes (defective or qualified). For example, while CLIP can easily and accurately identify a nut in the image, it may still treat a nut with cracks as simply a nut,

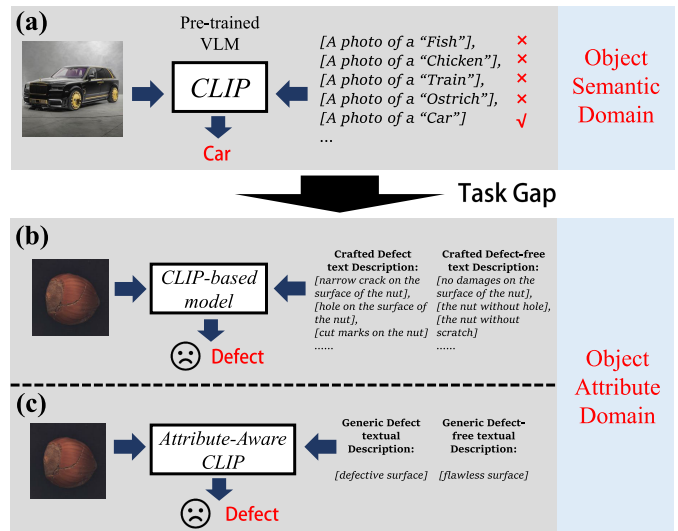


Fig. 1. Motivation of our research and the uniqueness of our proposed method. (a) Task for which the original CLIP is adapted, (b) complex crafted text prompts required by current language guided ZSDD methods, and (c) overview of our ZSDD method which only requires generic and simple text prompts.

without paying attention to the defect. Therefore, specific text descriptions of cracks must be designed to inject additional knowledge into CLIP to detect defects. Given the endless and unpredictable nature of defect appearance, this drawback of requiring manually designed text prompts severely hinders the model’s detection performance and practical value.

The key to solving the problem of requiring complex and tedious text prompts lies in bridging the domain gap between CLIP’s focus on object category semantics and the attribute discovery required by ZSDD. To this end, we propose a novel ZSDD framework called attribute-aware CLIP, which only requires simple and generic text prompts to perform cross-category ZSDD, such as “defective” and “normal.” The core of attribute-aware CLIP is a learnable textual domain bridge, which is designed to translate the simple text feature prototypes into prompt embeddings that are more aligned with attribute awareness. By translating text embeddings through the textual domain bridge, we decouple the original CLIP image–text matching process, reducing CLIP’s focus on object semantics and guiding the model to pay attention to the presence or absence of anomalies in the image. This allows for the transformation of encoded simple text feature prototypes into enriched prompt embeddings with strong attribute awareness, leading to more effective matching responses to defect information in the image feature space. Additionally, we first perform component clustering on the inspected image before feature encoding, which diminishes global semantic focus while enhancing the local image features. This encourages the model to further concentrate on abnormal attributes rather than object semantics.

Compared to current CLIP-based ZSDD methods, our attribute-aware CLIP has two main advantages: 1) as shown in the comparison between parts (b) and (c) of Fig. 1, our key feature is that it only requires simple, generic text prompts, without the need for special design. This advantage is highly significant for the generalization of the method and the rapid

application in real-world scenarios; and 2) additionally, the textual domain bridge we propose is a flexible and independent module. Unlike other methods that require fine-tuning the CLIP text encoder, it can be easily integrated into more advanced foundational VLMs in the future, enabling the rapid transfer of more powerful zero-shot capabilities to defect detection tasks.

To summarize, the main contributions of our paper are as follows.

- 1) We propose a novel ZSDD method called attribute-aware CLIP, which leverages transformed simple text prompt prototypes to perceive the attributes of the inspected image. Unlike current ZSDD methods that heavily rely on hundreds of meticulously designed manual text prompts, our approach significantly simplifies text prompt engineering and is effectively applicable to the inspection of various industrial product categories.
- 2) We design a textual domain bridge that can transform and enrich prompt embeddings with defect attribute perception capabilities within the text feature space. Compared to some current zero-shot detection methods that require fine-tuning the text encoder, our textual domain bridge approach is more flexible and straightforward, avoiding the risk of damaging the feature extraction capabilities of pretrained CLIP. Therefore, the textual domain bridge will enable any state-of-the-art (SOTA) VLMs to be directly applied to ZSDD tasks in the future.
- 3) We introduce the component clustering before the image encoder to reduce the global semantic features, thereby lessening CLIP's preference for object semantics and enabling a stronger focus on defect attribute discovery.
- 4) Experiments on the industrial datasets with 16 highly diverse semantic categories demonstrate that our method achieves outstanding ZSDD performance. Compared to current SOTA methods, we improve pixel-level AUROC by 2.14% and image-level AUROC by 2.42%.

The rest of this article is organized as follows. Section II reviews the related works to our study. Section III presents the details of the proposed method. Section IV reports our experimental setup, the results of ablation, and comparison experiments to demonstrate the effectiveness of the proposed method. Finally, Section V provides the conclusion and future work.

II. RELATED WORK

A. Data-Driven Defect Detection Methods

The data-driven methods have advanced visual defect detection by facilitating automatic feature extraction with deep learning techniques. Depending on whether data labels are used, these techniques can be classified into supervised learning defect detection methods and unsupervised learning defect detection methods.

1) Supervised Defect Detection: In early research, common object detection methods were applied to industrial defect detection. For instance, improved faster R-CNN was used to detect surface defects for the components with a small diameter and deep aperture [26], while the improved sin-

gle shot multibox detector (SSD) object detection network was adopted to locate the defect area of fastener on the catenary support device [27]. Subsequently, some methods specifically designed for defect detection tasks were developed. Zou et al. [28] propose a DeepCrack network based on encoder-decoder architecture for ground crack detection. The U-shaped deep residual neural network [29] is proposed for the conductive particle detection in the manufacturing process of thin film transistor liquid crystal display (TFT-LCD). Wen et al. [30] propose a feature pyramid network with atrous convolution (FPNAC) for defect feature fusion. Zhao et al. [1] propose an attention-based multiscale feature fusion (AMFF) module, aiming to enhance defect features and improve defect identification by leveraging the attention mechanism in the feature fusion. An improved YOLOv5 method [31] including the adaptive spatial feature fusion (ASFF) is proposed for improving defect detection accuracy. MFANet [32] is proposed for defect segmentation in seamless steel tubes, which learns the defect segmentation capability on the coarse-grained labeled defect dataset and generalizes the capability into fine-grained defective sample segmentation.

However, the scarcity of defect samples makes it challenging to accurately and adequately train supervised defect detection methods.

2) Unsupervised Defect Detection: Due to the difficulty in obtaining sufficient defect samples to meet the training requirements of supervised methods, unsupervised defect detection methods have garnered significant attention. Unsupervised defect detection methods only require defect-free samples for training. These methods typically model the distribution of normal samples and consider any deviation from this model as a defect. Reconstruction-based methods generate a reconstructed image of the inspected target and then detect defects using the reconstruction error between the input and the reconstruction. MemAE [33] proposes the memory mechanism that stores the prototypical normality patterns to mitigate the high-fidelity anomaly reconstruction problem encountered in conventional auto-encoders. DFR [34] leverages the local semantic perception capability of CNNs, enabling it to extract multilevel features and perform feature-level reconstruction, thus achieving more robust anomaly detection performance. PFM [35] employs feature mapping and correspondence mechanisms to establish different normal relationships across various feature domains, thereby extending the applicability of reconstruction-based methods.

Due to background noise disturbance and misalignment between pixels in the reconstructed and input images, reconstruction-based methods can suffer from detection errors. To address these challenges, researchers have turned their attention to the feature space, aiming to achieve more robust defect detection through high-dimensional feature similarity measures. SPADE [36] constructs a feature vector database of normal samples and uses k -nearest neighbors (kNN) technology to retrieve the k most semantically similar normal samples to the inspected sample. Finally, a feature pyramid matching method is utilized for multiscale feature alignment, enhancing the robustness of defect localization. Rippel et al. [37] extract multiscale features from normal samples using a pretrained

network. Then, they model each feature map as a multivariate Gaussian distribution. The Mahalanobis distance between the feature vectors of the inspected sample and the normal distribution is used to determine the defects. Recently, anomaly detection localization methods based on a normalized flow framework have been introduced. CS-Flow [12] is constructed based on the normalized flow framework. This method jointly processes multiscale feature maps and uses normalized flow to assign likelihoods to the input samples as the defect probability.

Although unsupervised defect detection effectively addresses the dependency on defect data for training, these data-driven methods can only perform defect detection for a single class according to the training data, and cannot handle cross-class defect detection. Additionally, the training time and computational resources required for these methods in practical applications are also significant factors that cannot be overlooked.

B. Zero-Shot Learning Based on VLMs

Recently, large pretrained VLMs [19], [38] have achieved remarkable success, with CLIP being the most renowned among them. CLIP [20] is the first model pretrained on a billion-scale image-text dataset, demonstrating impressive versatility. Language-driven zero-shot ability has effectively enhanced robustness for downstream tasks [39] and exhibited superior cross-modal semantic alignment. Subsequent VLM research have explored large-scale pretraining from various perspectives, such as more efficient model designs [40] and multitask joint training [41].

Visual community has demonstrated CLIP's zero-shot transfer capabilities across various downstream tasks. Open-vocabulary object detection is a common CLIP-based task. DetPro [22] learn continuous prompt representations for open-vocabulary object detection based on the pretrained VLM. Wu et al. [42] group contextually related regions into a bag. The embeddings of the regions within this bag are treated as the embeddings of words in a sentence and are then sent to the VLM's text encoder to obtain a region bag embedding. This embedding was learned to align with the corresponding features extracted by the frozen VLM. Additionally, other works [43], [44] have utilized pretrained CLIP models for language-guided segmentation and achieved good performance. Several studies have pointed out that prompt engineering and fine-tuning can significantly enhance CLIP's zero-shot generalization performance for specific tasks [20], [45]. WinCLIP [23] revealed that through carefully crafted and extensive prompt engineering, CLIP can effectively perform defect detection tasks. PromptAD [24] is another contemporary ZSDD work and also aims to adapt pretrained CLIP for ZSDD. AnoCLIP [46] particularly introduces a unified domain-aware contrastive state prompting template for fine-grained vision-language matching, which exploits the potential of CLIP for zero-shot defect localization.

However, the cumbersome prompt engineering in these works not only creates practical application barriers but also diminishes their performance in detecting defects related to unseen object semantics. Compared to the aforementioned

CLIP-based ZSDD methods, our attribute-aware CLIP introduces a flexible and independent textual domain bridge to transform generalized text prompts, enabling CLIP to perceive defect attributes. In other words, we can achieve ZSDD using only simple and generic text prompt embeddings, and our method outperforms the current SOTA methods.

III. PROPOSED METHODOLOGY

First, the overall framework of our ZSDD method is described in Section III-A. Next, we will detail the specifics of the textual domain bridge and component clustering of our method in Sections III-B and III-C, respectively. Finally, in Section III-D, we will explain how to train the textual domain bridge with auxiliary data and how to employ attribute-aware CLIP for zero-shot reference in defect detection tasks.

A. Overall Pipeline

In this article, we propose attribute-aware CLIP, motivated by reducing the expertise, comprehensiveness, and complexity of text prompts required for ZSDD in existing methods. The core principle for defect detection behind attribute-aware CLIP's approach is to use two types ("defective" and "non-defective") of text prompt embeddings to calculate the alignment between these embeddings and the features of the inspected image, thereby indicating the presence or absence of defects.

As illustrated in Fig. 2, we first use CLIP's text encoder to encode the generic text prompts, forming text feature prototypes. Next, the textual domain bridge, the core of attribute-aware CLIP, translates and enriches the text feature prototypes into generalized prompt embeddings for defective and nondefective classes, endowing these embeddings with the ability to perceive attributes of the image features. To further enhance CLIP's focus on local attributes and reduce its attention to the overall semantics of the object, we designed a unique stage in the processing branch of the inspected image, called component clustering. This stage breaks down the entire image into components of the industrial product. In CLIP's visual encoder, we fuse global image features with component features at multiple intermediate layers to extend more details of local visual features, forming the final visual embedding (used to obtain the defect score) and the local visual embedding (used to locate defect regions). The simplified ZSDD process of our attribute-aware CLIP is shown in Algorithm 1, and the detailed process of each step will be introduced in Sections III-B–III-D.

B. Textual Domain Bridge

Current language-guided ZSDD methods must incorporate prior semantic knowledge of defect descriptions into text prompts, such as "*narrow crack on the surface of the [obj]*," to obtain text embeddings to query corresponding visual embeddings. It is obvious that these manually crafted text prompts cannot cover all possible defect types and require expert input for prompt design. Therefore, the central focus of our work is how to utilize generic defect-related text prompts,

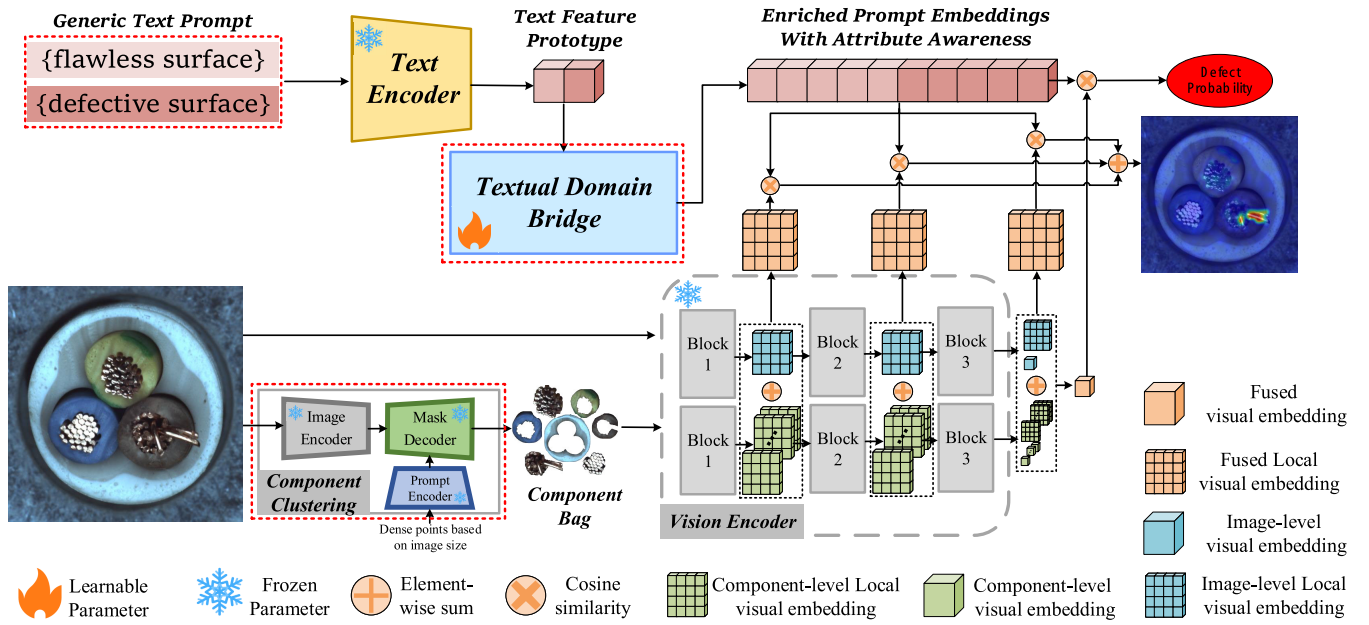


Fig. 2. Overview of our attribute-aware CLIP. The textual domain bridge is the core of attribute-aware CLIP, transforming simple and generic text feature prototypes into enriched prompt embeddings with attribute awareness (Section III-B). Component clustering is another proposed unique module that reduces global semantic information in images, encouraging CLIP to focus on local defect attributes (Section III-C).

Algorithm 1 Zero-Shot Defect Detection of Attribute-Aware CLIP

Input: Inspected Image $\mathcal{I}$, Generic Defect Text Prompt T_d , Generic Normal Text Prompt T_n ;

Functions: Text Encoder E_{text} , Visual Encoder E_{visual} , Textual Domain Bridge f_{TDB} , Component Clustering CC , feature fusion $Fuse$, similarity calculation Sim ;

Outputs: Defect Localization Map Map , Defect Probability $Prob$;

$Proto = \text{concat}(E_{text}(T_n), E_{text}(T_d))$

$Emb_{prompt} = f_{TDB}(Proto)$

$Comp = CC(\mathcal{I})$

$Emb_{visual}, Emb_{visual}^{local} = Fuse(E_{visual}(\mathcal{I}), E_{visual}(Comp))$

$Prob = Sim(Emb_{visual}, Emb_{prompt})$

$Map = Sim(Emb_{visual}^{local}, Emb_{prompt})$

such as “defective surface,” to query visual embeddings effectively. However, CLIP’s original pretraining is aligned with object semantics rather than the defects and normalcy within images. To address this task domain gap, we designed a textual domain bridge, which can be viewed as a translator that converts simple and generic text feature prototypes into enriched attribute-aware prompt embeddings. In other words, the primary purpose of the textual domain bridge is to transform generic text embeddings (e.g., “flawless surface” and “defective surface”) into prompt embeddings that are better aligned with attribute awareness. This transformation bridges the gap between CLIP’s original design focus on object category semantics and the attribute discovery required by the ZSDD task. This whole transformation process can be defined as

$$P = \text{Concat}(EC_T(T_n), EC_T(T_d)) \quad (1)$$

$$E = f_{TDB}(P) \quad (2)$$

where EC_T represents the text encoder, T_n and T_d denote the generic text prompts for nondefective and defective cases, respectively, P refers to the text feature prototype, f_{TDB} represents the mapping function of the textual domain bridge, and E stands for the enriched prompt embeddings with attribute awareness.

As shown in Fig. 3, the textual domain bridge employs an encoder-decoder structure for text semantic expansion, followed by a fully connected layer for dimensional adjustment, ensuring that the transformed prompt embeddings match the dimensions of the image embeddings produced by the visual encoder for subsequent similarity calculations. Specifically, within each encoder and decoder, a long short-term memory (LSTM) network is used as the foundational module. The text feature prototypes are treated as a long sequence $P = \{p_1, p_2, \dots, p_T\}$. We first use an encoder for sequence integration and semantic compression of the text feature prototypes to form intermediate context vectors which contain the information of the input sequence. The encoding process can be expressed mathematically as follows:

$$\mathbf{h}_t^{(e)} = \text{LSTM}(\mathbf{p}_t, \mathbf{h}_{t-1}^{(e)}) \quad (3)$$

$$\mathbf{V}_{\text{context}} = \mathbf{h}_T^{(e)} \quad (4)$$

where $\mathbf{h}_t^{(e)}$ is the hidden state at time step t , $\mathbf{h}_{t-1}^{(e)}$ is the previous hidden state, and the final hidden state of the encoder $\mathbf{h}_T^{(e)}$ is used as the context vector $\mathbf{V}_{\text{context}}$.

Then, we use the decoder to decompress the intermediate context vectors and embed attribute-aligned knowledge. Using the LSTM structure, we take the intermediate context vector as the guide for generating each embedding. This design ensures that the enriched prompt embeddings gain attribute awareness while still staying true to the input semantics of the text feature prototypes. Finally, the decoder’s hidden state $\mathbf{h}_t^{(d)}$ is

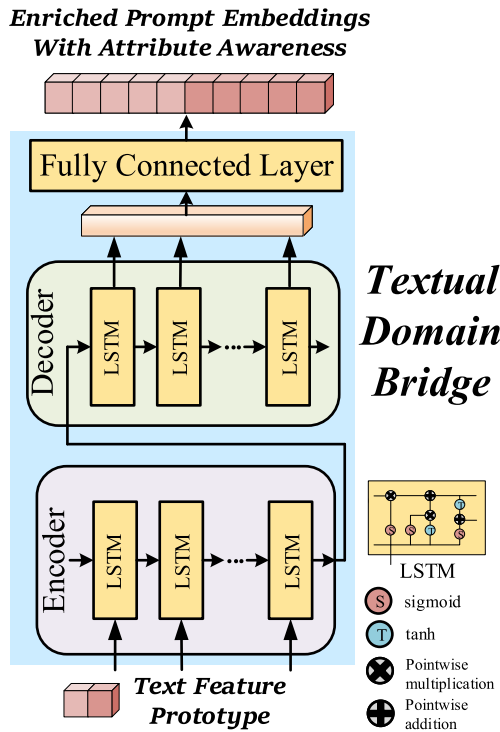


Fig. 3. Detail structure of the textual domain bridge.

passed through a fully connected layer to produce the output $\mathbf{y}_t$. The decoding process iteratively generates the output sequence using the LSTM in conjunction with the intermediate context vector and the previously generated outputs. Finally, the transformed long sequence $E = \{e_1, e_2, \dots, e_T\}$ is the enriched prompt embeddings with attribute awareness. The whole decoding process can be depicted as follows:

$$\mathbf{h}_0^{(d)} = \mathbf{V}_{\text{context}} \quad (5)$$

$$\mathbf{h}_t^{(d)} = \text{LSTM}(\mathbf{e}_{t-1}, \mathbf{h}_{t-1}^{(d)}) \quad (6)$$

$$\mathbf{e}_t = \text{FC}(\mathbf{h}_t^{(d)}) \quad (7)$$

where $\mathbf{h}_t^{(d)}$ is the hidden state of the decoder at time step t , $\mathbf{e}_{t-1}$ is the output from the previous time step, and FC is the function of fully connected layer. The training of learnable parameters within the textual domain bridge is distinct from typical supervised learning, which relies on ground truth corresponding to the current data modality. Instead, we design a cross-modal learning approach that utilizes image information of auxiliary data for decoupled attribute knowledge injection. The detailed method will be discussed in Section III-D. Due to the simplicity of our textual domain bridge, we only require one-tenth of the auxiliary data size for learning compared to other ZSDD methods that require fine-tuning the text encoder.

Importantly, our textual domain bridge is an independent and flexible module. It can be directly cascaded after the text encoder to perform prompt embedding transformation without requiring fine-tuning of the VLM's text encoder. As a result, our textual domain bridge will allow any future SOTA image-text models to be directly applied to the ZSDD task. This flexibility and modularity are key differentiators compared to other CLIP-based methods for ZSDD.

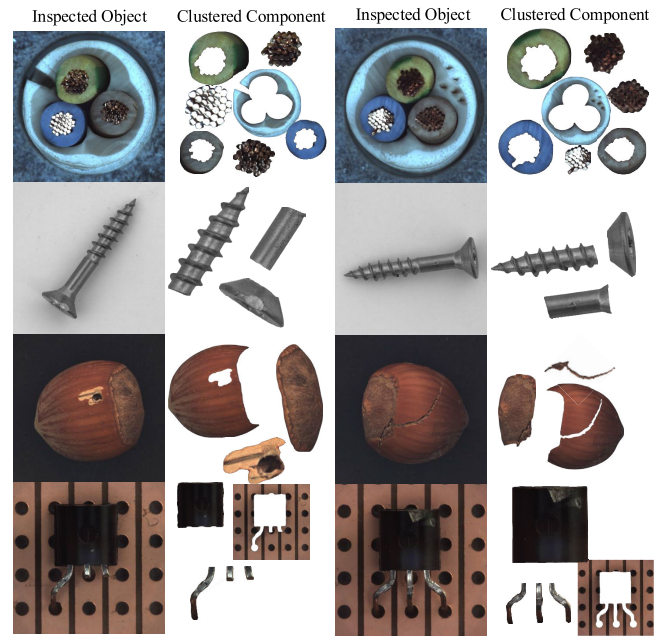


Fig. 4. Examples of the component clustering results.

C. Component Clustering

Component clustering is a unique image preprocessing stage in our attribute-aware CLIP, compared with other ZSDD methods based on VLM. The purpose of this stage is to segment a complete object into multiple independent parts, with each part potentially representing different regions of the object. These parts are often also the complete component expression of the object. Then, they will be fed into the visual encoder one by one and fused with the global visual features.

In industrial defect detection, different parts of an object are clustered separately, which can shift the focus from the global image to local areas. This operation reduces CLIP's attention to the overall semantics of the target and stimulates the potential of attribute-aware capabilities. Moreover, clustering based on components can make it easier to detect subtle defects, such as cracks, scratches, or wear.

In practical implementation, we use an open-source SAM [47] component (sam_vit_h) as the basis. The detailed structure is illustrated in the lower-left part of Fig. 2 within the "Component Clustering" box. Based on the input size of the inspected image, uniform prompt points are automatically set for SAM input, and post-processing is used to remove stray pixels and holes.

Fig. 4 shows some examples of the component clustering results. Each clustered component is independently segmented, highlighting which regions are grouped together. This clustering helps the visual model focus on defects in these local areas without being distracted by global features.

D. Zero-Shot Reference and Cross-Modal Attribute Awareness Injection With Auxiliary Data

1) **Zero-Shot Reference:** We transform two types of generic text prompt prototypes (i.e., defect and nondefect) into corresponding text embeddings with attribute awareness E_d and E_n via the textual domain bridge. The image encoder of

CLIP encodes the input image I to derive visual features, where the class token $f_I \in \mathbb{R}^D$ is the visual embedding, and the patch tokens $f_I^P \in \mathbb{R}^{H \times W \times D}$ are the local visual embeddings. Attribute-Aware CLIP measures the similarity between the text and fused visual embedding for ZSDD. Specifically, the probability that image I is defective can be expressed by the following formula:

$$P(E_d, f_I) = \frac{e^{(\cos(E_d, f_I)/\tau)}}{e^{(\cos(E_d, f_I)/\tau)} + e^{(\cos(E_n, f_I)/\tau)}} \quad (8)$$

where τ is the temperature hyperparameter, set to 0.07 in our implementation, and $\cos()$ denotes the cosine similarity calculation. For pixel-wise prediction, the fused visual embedding is changed by the fused local visual embeddings to generate the corresponding localization maps $M_n \in \mathbb{R}^{H \times W}$ and $M_d \in \mathbb{R}^{H \times W}$, where each element (j, k) is calculated as $P(E_n, f_{I(j,k)}^P)$ and $P(E_d, f_{I(j,k)}^P)$. The localization maps M_n and M_d from all intermediate layers are combined and the interpolation and smoothing operations to obtain the final defect localization map.

2) Cross-Modal Attribute Awareness Injection With Auxiliary Data: To avoid the incompleteness of manually crafted prompts, we use the defect or nondefect information expressed by image features as the ground truth for the textual domain bridge outputs. To effectively inject attribute-aware knowledge into the textual domain bridge, we employ joint alignment optimization using both global and local visual information. This use of visual information to supervise the text modality is a decoupled optimization approach that effectively disrupts the original pretrained CLIP's focus on image object semantics. Specifically, we calculate the loss using adapted InfoNCE loss for global alignment and use focal loss [48] for local alignment. The final complete loss function is expressed as follows:

$$L_{\text{global}} = -\log \frac{\exp(f_I \cdot E_+/\tau)}{\exp(f_I \cdot E_d/\tau) + \exp(f_I \cdot E_n/\tau)} \quad (9)$$

$$L_{\text{local}}^{i \in l} = \text{Focal}(\text{Up}(\text{concat}(M_d^i, M_n^i)), M_{\text{GT}}) \quad (10)$$

$$\text{Loss} = L_{\text{global}} + \lambda \sum_i L_{\text{local}}^i \quad (11)$$

where E_+ represents the text embedding that correctly matches the image, l represents the set of intermediate layers used, M_{GT} is the ground-truth mask of the image, Focal represents the focal loss, Up means the upsampling operation, concat is concatenation along with the channel, and λ is a hyperparameter to adjust the global and local information supervision.

Notably, compared to other ZSDD methods that require retraining the CLIP text encoder with auxiliary data, our attribute-aware CLIP fully retains the text encoder's semantic extraction capabilities, requiring only the optimization of the learnable parameters within the text domain bridge. As a result, the size of auxiliary data needed for our method is merely one-tenth of that required by other methods.

IV. EXPERIMENTAL RESULTS

We conduct an array of experiments to evaluate the ZSDD performance of attribute-aware CLIP, focusing on both defect discovery and localization. Additionally, we perform extensive ablation studies to validate the effectiveness of our proposed

components. In this section, we will provide a detailed overview of the datasets used, evaluation metrics, implementation details, comparison of visualized defect localization results among different methods, quantitative comparisons of detection performance, and the ablation studies.

A. Dataset Description

To better demonstrate the superior performance and practical value of our method, we select recent challenging real-world defect datasets covering various defect detection scenarios. Specifically, we evaluate a total of 16 product categories from the MVTecAD [49], KolektorSDD2 [50], and VisA [51] datasets.

1) MVTecAD Dataset: The MVTecAD dataset is a widely used and authoritative dataset for defect detection, designed to provide a benchmark for algorithms in the field of industrial visual defect detection. It contains high-resolution images across 15 different categories, covering a variety of real-world scenarios. The dataset includes more than 5000 images, with sizes ranging from 840×840 to 1240×1240 pixels. The collected defects vary in size, shape, and type, totaling 70 different types of defects. MVTecAD also provides pixel-level ground truth annotations for each defect sample, enabling the evaluation of both detection accuracy (whether a defect is present) and localization performance (the position of the defect within the image). Therefore, this dataset allows for a comprehensive assessment of ZSDD performance for our method and other SOTA methods.

2) KolektorSDD2 Dataset: The KolektorSDD2 defect dataset is derived from images of industrial manufacturing processes, provided and annotated by Kolektor Group, including 356 defective images and 2979 nondefective images. The images have a resolution of approximately 230×630 pixels. This dataset is characterized by its focus on real defect scenarios, where defects may be subtle and challenging to detect, including scratches, dents, and other surface irregularities. KolektorSDD2 also provides annotated images with precise defect localization, allowing for the evaluation of detection accuracy and defect localization performance. Since KolektorSDD2 reflects real-world challenges, it is often used as a benchmark dataset to evaluate the reliability and accuracy of defect detection algorithms.

3) VisA Dataset: The VisA dataset is a high-resolution benchmark dataset focused on defect detection and localization in industrial scenarios. It contains high-resolution images across 12 different categories. It also includes diverse defect types (such as scratches, contamination, missing parts, and deformation) with pixel-level annotations, enabling precise evaluation of detection and localization. It is highly suitable for ZSDD scenarios and serves as a valuable resource for developing new ZSDD algorithms.

B. Evaluation Indicators

We comprehensively evaluate defect detection performance using four metrics. For detection accuracy, we report the area under the receiver operating characteristic curve (AUROC) and average precision (AP). For defect localization, we report

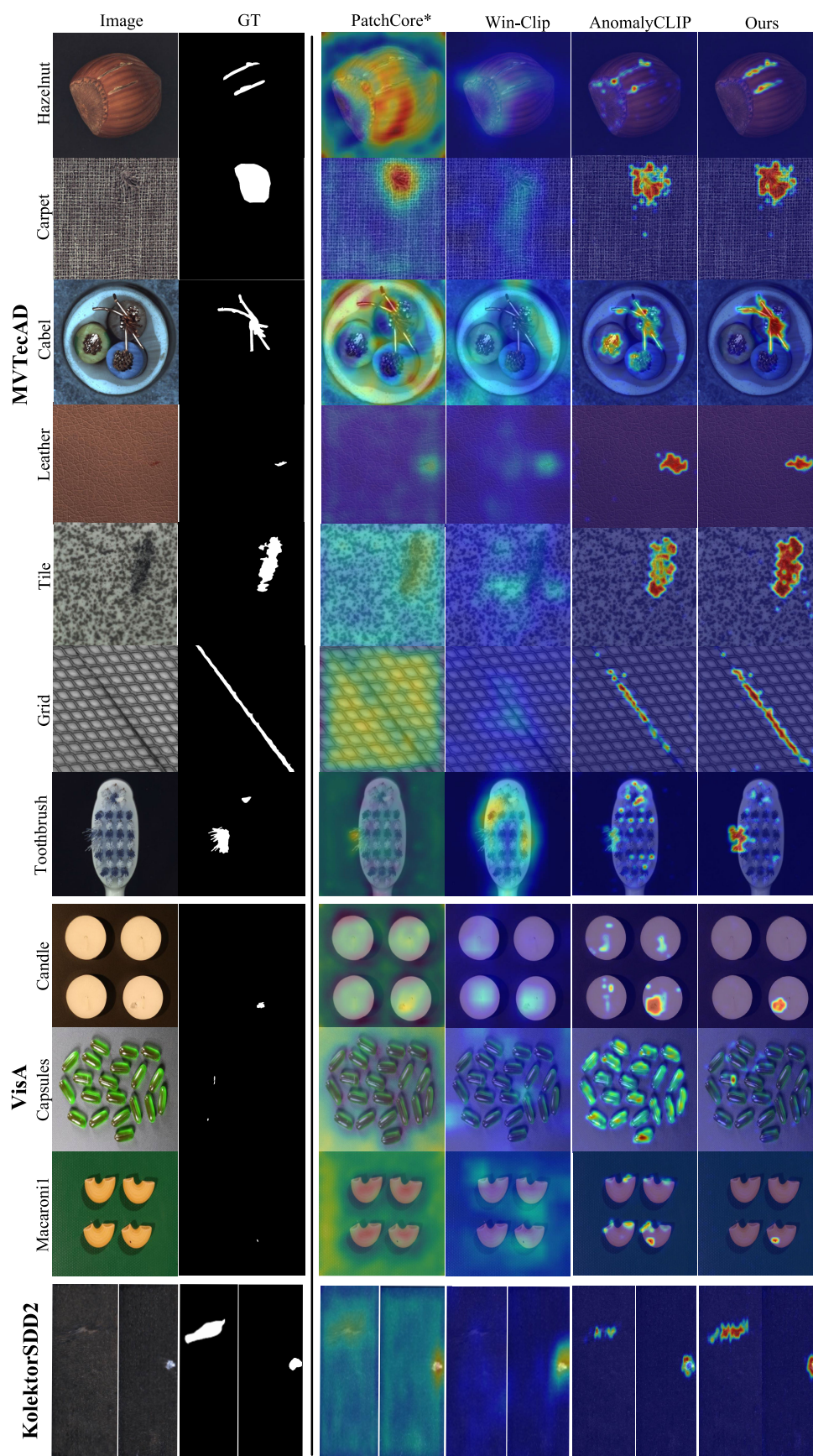


Fig. 5. Qualitative comparison of defect localization results on MVTec-AD, KolekorSDD2, and VisA dataset with different methods. * indicates training with only one sample.

pixel-wise AUROC (pAUROC), AUPRO, and AP [52], [53]. We aim to provide a deeper insight into the performance of various algorithms through the detection AP, localization AUPRO, and AP metrics.

C. Implementation Details

We adopt the open-source CLIP model (ViT-L/14) of OpenCLIP as our backbone. All pretrained model parameters are frozen. We implement our models in Python 3.10 and PyTorch 1.13.1. All experiments are conducted on the same Nvidia RTX 3090 GPU on Ubuntu 18.04 Linux. In our zero-shot experiments, we select a single category of data (e.g., cable, grid, etc.) as auxiliary data for learning, and test on the remaining 15 categories. We choose Adam with a 0.001 learning rate as an optimizer. The hyperparameter γ is 1.0 of the loss function in our implementation. For the fusion ratio between component-level embeddings and image-level embeddings, we set it to 0.19: 1. For smoothing operations in the final defect localization map, we choose a Gaussian filter with $\theta = 4$.

D. Visualized Defect Localization Results

In this experiment, we present the visualized defect localization results of 11 objects from three datasets. The compared methods include the advanced data-driven defect detection method PatchCore [54], the milestone ZSDD method WinCLIP [23], and the current SOTA ZSDD method AnomalyCLIP [53]. The heatmaps shown in the figure reflect each method's response to defect locations.

As illustrated in Fig. 5, our method generally demonstrates more precise and sensitive defect capture and localization capabilities. For defects with prominent anomalous features, such as those in the cable, carpet, and hazelnut columns, our method's defect localization results are more complete and comprehensive, with fewer noisy responses compared to prior methods. For challenging defects where the anomalous features closely resemble background features, such as those in the tile, leather, and KolektorSDD2 columns, our method shows a more sensitive and accurate localization ability than the other methods. In addition, Fig. 6 presents more visualized defect localization results, demonstrating the effectiveness of our method in defect detection for each category of the MVTecAD and VisA datasets.

E. Quantitative Defect Detection Results

We conduct quantitative comparison experiments on 16 target categories across two datasets to demonstrate the superior performance of our method in terms of both detection accuracy and defect localization. The compared SOTA methods include the base model CLIP [20] (2021 ICML), CLIP-AC [20] (2021 ICML), WinCLIP [23] (a ZSDD method that uses extensive fine-grained text descriptions) (2023 CVPR), VAND [55] (an improved version of WinCLIP) (2023 CVPR workshop), and the latest AnomalyCLIP method [53] (2024 ICLR).

1) *Comparison Results of Detection Accuracy: Tables I–IV* present the ZSDD accuracy of our attribute-aware CLIP compared to five competing methods on the MVTecAD, KolektorSDD2 and VisA dataset using image-level AUROC and

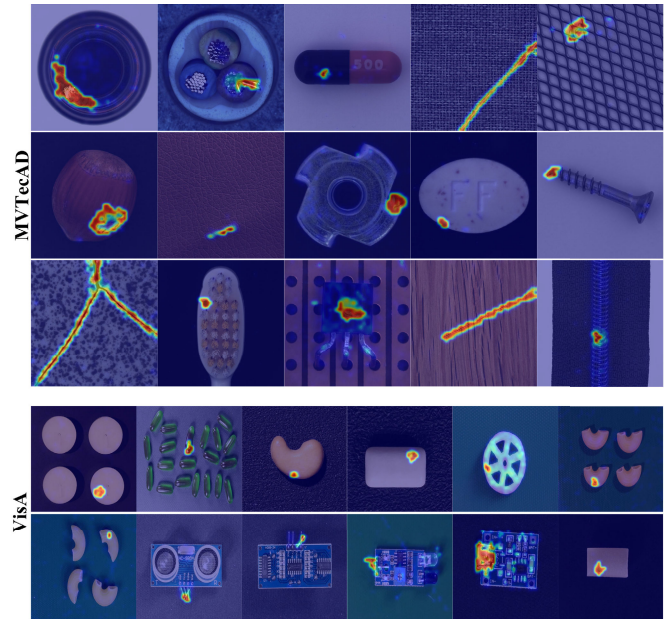


Fig. 6. Visualization of defect localization results for each category in the MVTecAD and VisA datasets.

TABLE I
PERFORMANCE COMPARISON (AUROC) FOR DETECTION
ACCURACY ON MVTecAD

AUROC	CLIP	CLIP-AC	Win-CLIP	VAND	Anomaly CLIP	Our method
carpet	96.0	93.1	100	99.5	100	100
bottle	45.9	46.1	99.2	92.0	89.3	90.1
hazelnut	88.7	91.1	93.9	89.6	97.2	97.4
leather	99.4	99.5	100	99.7	99.8	100
cable	58.1	46.6	86.5	88.4	69.8	93.2
capsule	71.4	68.8	72.9	79.9	89.9	90.0
grid	72.5	63.7	98.8	86.3	97.0	97.6
pill	73.6	73.8	79.1	80.5	81.8	82.7
transistor	48.8	51.2	88.0	80.8	92.8	95.0
metal_nut	62.8	63.4	97.1	68.4	93.6	93.1
screw	78.2	66.7	83.3	84.9	81.1	87.8
toothbrush	73.3	89.2	88.0	53.8	84.7	87.1
zipper	60.1	36.1	91.5	89.6	98.5	95.9
tile	88.5	89.0	100	99.9	100	100
wood	94.0	94.9	99.4	99.0	96.8	99.5
Mean	74.1	71.5	91.8	86.1	91.5	93.9

AP metrics. These datasets feature very different foreground objects, backgrounds, and defect types. Attribute-Aware CLIP achieves outstanding ZSDD performance across both datasets, significantly outperforming the other five methods in detection accuracy for most product categories. The poor performance of CLIP and CLIP-AC is due to the fact that CLIP's original pretraining focuses on matching object semantics rather than defect semantics. WinCLIP and VAND obtain better results using rich and fine-grained handcrafted text prompts. However, as previously mentioned, the labor-intensive creation of these text prompts limits their practical application and generalization. AnomalyCLIP utilizes learnable text prompts and retraining of the CLIP text encoder to also achieve excellent ZSDD performance. In contrast, our attribute-aware CLIP uses a text-domain bridge for semantic translation and

TABLE II
PERFORMANCE COMPARISON (AP) FOR DETECTION
ACCURACY ON MVTecAD

AP	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly- CLIP	Our method
carpet	98.8	97.8	100	99.8	100	100
bottle	78.9	79.8	99.8	97.7	97.0	96.9
hazelnut	94.6	95.9	96.9	94.8	98.6	98.3
leather	99.8	99.8	100	99.9	99.9	100
cable	70.8	64.3	91.2	93.1	81.4	95.0
capsule	92.1	90.9	91.5	95.5	97.9	98.0
grid	87.1	83.9	99.6	94.9	99.1	99.6
pill	93.4	93.6	95.7	96.0	95.4	94.6
transistor	48.1	49.9	87.1	77.5	90.6	93.3
metal_nut	87.7	89.2	99.3	91.9	98.5	99.4
screw	91.4	86.6	93.1	93.6	92.5	93.2
toothbrush	90.7	96.0	95.6	71.5	93.7	93.6
zipper	87.4	73.9	97.5	97.1	99.6	98.2
tile	95.9	96.2	100	100	100	100
wood	97.9	98.3	99.8	99.7	99.2	99.8
<i>Mean</i>	<i>87.6</i>	<i>86.4</i>	<i>96.5</i>	<i>93.5</i>	<i>96.2</i>	<i>97.3</i>

TABLE III
PERFORMANCE COMPARISON (AUROC AND AP) FOR DETECTION
ACCURACY ON KOLEKTORSDD2

Kolektor SDD2	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
AUROC	56.2	57.4	85.5	86.3	91.0	93.3
AP	49.8	50.9	76.2	80.0	82.1	90.5

TABLE IV
PERFORMANCE COMPARISON (IMAGE-LEVEL AUROC AND AP AND
PIXEL-LEVEL AUROC AND AUPRO AND AP) FOR DEFECT
DETECTION ACCURACY AND LOCALIZATION ON VISA

Metrics	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
AUROC	66.4	65.0	78.1	78.0	82.1	84.1
AP	71.5	70.1	81.2	81.4	85.4	85.9
pAUROC	46.6	47.8	79.6	94.2	95.5	95.5
AUPRO	14.8	17.3	56.8	86.8	87.0	71.7
pAP	1.2	1.2	5.3	25.7	21.3	30.4

enrichment of general text prototypes, which demonstrates the best performance. Specifically, on the MVTecAD dataset, our method outperforms the best existing method, WinCLIP requiring extensive handcrafted prompts, by a mean AUROC of 2.1%. Across the 15 subcategories of MVTecAD, our method achieves the best performance on 11 datasets, demonstrating its robustness and adaptability across various defect types. On the KolektorSDD2 dataset, our method surpasses the best existing method with an AUROC increase of 2.3%. Even on the challenging VisA dataset, our approach is still the best performance in both AUROC and AP metrics.

2) Comparison Results of Defect Localization: Tables IV–VIII present the quantitative defect localization performance across 28 product categories, comparing our method to five others. As demonstrated in the visualized defect localization results in Fig. 5, our attribute-aware CLIP exhibits more precise defect localization capabilities. Specifically, according to the MVTecAD dataset, our method achieves

TABLE V
PERFORMANCE COMPARISON (PIXEL-LEVEL AUROC) FOR
DEFECT LOCALIZATION ON MVTecAD

pAUROC	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
carpet	11.5	10.7	95.4	98.4	98.8	99.2
bottle	17.5	23.3	89.5	83.4	90.4	90.8
hazelnut	25.2	34.0	94.3	96.1	97.1	97.2
leather	9.9	5.6	96.7	99.1	98.6	98.6
cable	37.4	37.5	77.0	72.3	78.9	95.7
capsule	50.9	49.1	86.9	92.0	95.8	96.6
grid	8.7	11.9	82.2	95.8	97.3	98.5
pill	55.8	60.8	80.0	76.2	92.0	90.7
transistor	51.1	48.5	74.7	62.4	71.0	73.4
metal_nut	43.9	53.6	61.0	65.4	74.4	78.2
screw	80.1	76.4	89.6	97.8	97.5	97.6
toothbrush	36.3	35.0	86.9	95.8	91.9	96.0
zipper	51.5	44.7	91.6	91.1	91.4	93.1
tile	49.9	39.1	77.6	92.7	94.6	96.5
wood	45.7	42.4	93.4	95.8	96.5	96.6
<i>Mean</i>	<i>38.4</i>	<i>38.2</i>	<i>85.1</i>	<i>87.6</i>	<i>91.1</i>	<i>93.2</i>

TABLE VI
PERFORMANCE COMPARISON (AUPRO) FOR DEFECT
LOCALIZATION ON MVTecAD

AUPRO	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
carpet	2.9	1.9	84.1	48.5	90.1	97.2
bottle	1.4	4.9	76.4	45.6	80.9	80.6
hazelnut	2.8	9.4	81.6	70.3	92.4	92.9
leather	0.2	0.0	91.1	72.4	92.2	96.0
cable	7.3	6.9	42.9	25.7	64.4	90.5
capsule	13.2	14.9	62.1	51.3	87.2	90.6
grid	0.9	2.4	57.0	31.6	75.6	94.5
pill	6.0	8.2	65.0	65.4	88.2	88.7
transistor	15.3	11.2	43.4	21.3	58.1	63.7
metal_nut	2.9	10.3	31.8	38.4	71.0	75.2
screw	57.8	56.2	68.5	67.1	88.0	88.8
toothbrush	5.8	5.2	67.7	54.5	88.5	89.2
zipper	17.7	15.2	71.7	10.7	65.3	77.3
tile	21.5	16.3	51.2	26.7	87.6	88.5
wood	13.7	10.3	74.1	31.1	91.2	93.2
<i>Mean</i>	<i>11.3</i>	<i>11.6</i>	<i>64.6</i>	<i>44.0</i>	<i>81.4</i>	<i>87.1</i>

an average improvement of 8.1% over WinCLIP in pixel-level AUROC. Among the 15 subcategories, our method achieves the highest performance on 11 datasets, showcasing its ability to localize anomalies effectively. Compared to AnomalyCLIP, our method also outperforms it by 2.14% in average pixel-level AUROC, again highlighting the superior performance of our method in locating defects. The AUPRO and pixel-level AP metrics also reflect the superior defect localization performance of our method on MVTecAD and KolektorSDD2 datasets. Even on the challenging VisA dataset, the average pixel-level AP metric of our method is still 4.7% higher than the second-best method. This is due to the attribute-aware text prompt embeddings, which have a more sensitive response to defect features, combined with the fusion of multilayer image-level local visual embeddings and component-level local visual embeddings.

TABLE VII

PERFORMANCE COMPARISON (PIXEL-LEVEL AP) FOR DEFECT LOCALIZATION ON MVTecAD

pAP	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
carpet	2.4	2.1	24.3	67.5	56.6	70.0
bottle	1.7	3.9	49.0	53	55.3	70.2
hazelnut	1.9	4.2	35.4	49.7	43.4	57.2
leather	0.3	0.1	23.7	52.3	22.7	25.1
cable	5.9	4.4	3.0	18.2	12.3	47.5
capsule	9.9	10.3	5.8	29.7	27.7	32.4
grid	0.4	1.7	3.0	36.6	24	37.8
pill	6.3	9.0	18.9	23.6	34.1	44.1
transistor	11.2	8.8	12.5	11.7	15.6	17.4
metal_nut	3.0	8.5	9.9	25.9	26.4	34.3
screw	29.9	31.5	3.1	33.7	27.5	35.1
toothbrush	4.1	4.2	6.6	43.2	19.4	54.7
zipper	20.0	19.3	20.5	38.7	38.7	39.4
tile	24.2	20.6	17.1	66.3	61.7	67.9
wood	10.7	12.3	33.6	61.8	52.7	66.0
Mean	8.8	9.4	17.8	40.8	34.5	46.6

TABLE VIII

PERFORMANCE COMPARISON (PIXEL-LEVEL AUROC AND AUPRO AND AP) FOR DEFECT LOCALIZATION ON KOLEKTORSDD2

Kolektor SDD2	CLIP	CLIP -AC	Win- CLIP	VAND	Anomaly CLIP	Our method
pAUROC	9.9	10.5	77.9	77.4	78.9	97.7
AUPRO	0.9	1.8	70.1	67.1	72.6	91.9
pAP	1.1	1.3	18.7	51.3	36.0	59.1

TABLE IX

IMPACT OF TEXTUAL DOMAIN BRIDGE ON DETECTION ACCURACY AND DEFECT LOCALIZATION

Module	MVTecAD		KolektorSDD2	
Tex. Do. Bri.	AUROC	p-AUROC	AUROC	p-AUROC
✓	93.9	93.2	93.3	97.7
✗	80.6 ↓13.3	78.0 ↓15.2	70.9 ↓22.4	68.5 ↓29.2

F. Ablation Studies

1) *Impact of Removing Textual Domain Bridge*: In Table IX, we report the individual effect of the textual domain bridge on enhancing ZSDD performance. When the textual domain bridge is removed, both detection accuracy and defect localization performance experience a significant decline. This further emphasizes that ordinary general text prompt embeddings have weak responses to global and local defect features, while the prompt embeddings empowered by our textual domain bridge exhibit outstanding attribute awareness.

2) *Impact of Removing Component Clustering*: Table X shows that the removal of the component clustering operation results in a noticeable decline in ZSDD performance, strongly validating the effectiveness of component clustering. Compared with ablation results on MVTecAD, the minimal impact of component clustering on the performance in the KolektorSDD2 dataset is primarily due to the nature of KolektorSDD2, which features background-free surface defects without component semantics (as shown in the far-right column of Fig. 5), limiting the effectiveness of component

TABLE X

IMPACT OF COMPONENT CLUSTERING ON DETECTION ACCURACY AND DEFECT LOCALIZATION

Module	MVTecAD		KolektorSDD2	
Comp. Clust.	AUROC	p-AUROC	AUROC	p-AUROC
✓	93.9	93.2	93.3	97.7
✗	90.6 ↓3.3	90.9 ↓2.3	92.4 ↓0.9	97.1 ↓0.6

TABLE XI

IMPACT OF PRETRAINED MODEL SELECTION ON DETECTION ACCURACY AND DEFECT LOCALIZATION

Pre-trained Model	MVTecAD		KolektorSDD2	
	AUROC	p-AUROC	AUROC	p-AUROC
ViT-L/14	93.9	93.2	93.3	97.7
ViT-B/16	95.3	94.5	84.2	94.8
ViT-B/32	89.1	84.9	80.6	70.4

clustering. This outcome aligns with our expectations regarding the effect of component clustering.

3) *Impact of Pretrained Model*: Table XI illustrates the impact of using different pretrained models on detection performance. Overall, ViT-B/32 exhibits the worst performance across various metrics and datasets. ViT-L/14 and ViT-B/16 demonstrate mixed results in terms of detection accuracy and defect localization capabilities on different datasets. However, ViT-B/16 shows a significant drop in image-level AUROC, which reflects detection accuracy, on the KolektorSDD2 dataset. Therefore, in practical deployments, our method can flexibly select the most suitable pretrained model based on the specific characteristics of the target detection task.

V. CONCLUSION AND DISCUSSION

In this article, we propose a novel ZSDD method, attribute-aware CLIP, which adapts CLIP to the discovery of defect attributes. Our approach addresses the shortcomings of current ZSDD methods that require extensive fine-grained textual prompts, resulting in high usage costs and limited generalization. The textual domain bridge of attribute-aware CLIP transforms and enriches simple yet generic descriptions of defective and nondefective conditions into attribute-aware prompt embeddings, enabling attribute-aware CLIP to perform unseen defect detection and localization. Additionally, to further direct CLIP's focus toward local defect attributes, we introduce component clustering in the image preprocessing stage to break down global object semantics in the image, thereby enhancing local visual features. Extensive experimental results on 16 public datasets demonstrate that attribute-aware CLIP achieves SOTA ZSAD performance.

We would like to specifically highlight that the textual domain bridge in attribute-aware CLIP is a simple yet independent module. This module is not embedded within the original text encoder and does not require fine-tuning the text encoder of VLMs. In practice, the textual domain bridge only needs to be cascaded directly after the text encoder to perform prompt embedding transformation. As a result, the textual domain bridge enables any future SOTA visual-language model to be adapted to ZSDD tasks, which is of great significance.

We have also observed that the selection of auxiliary datasets for injecting attribute-aware ability into the prompt embeddings can indeed affect the detection performance variance between different categories. To address this in practical algorithm deployment, we recommend utilizing a multimodal voting mechanism to effectively mitigate this impact. In future research, our goal is to build a large-scale industrial domain dataset for large-scale pretraining of the text domain bridge, further enhancing the ZSDD performance of our framework.

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